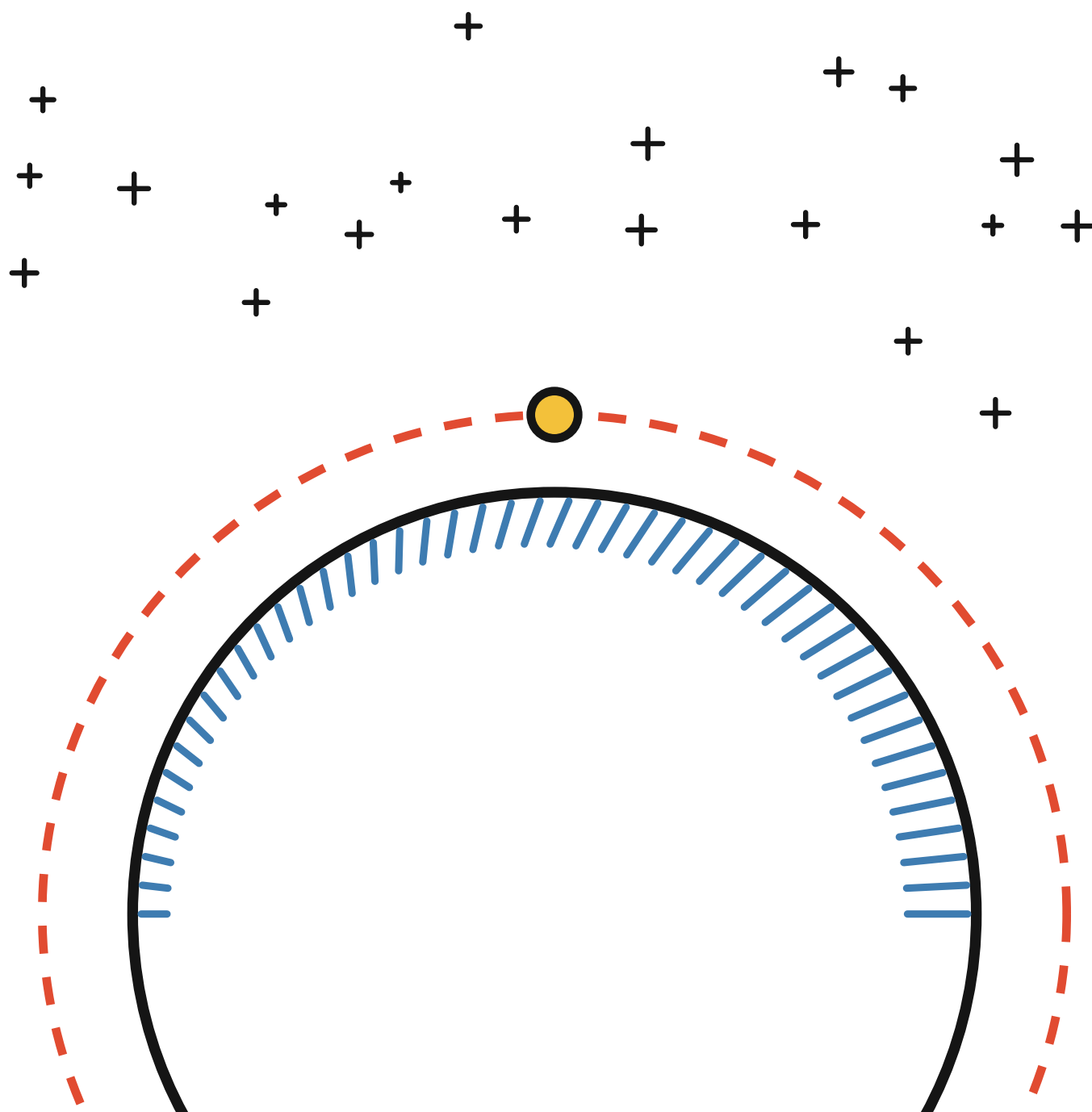


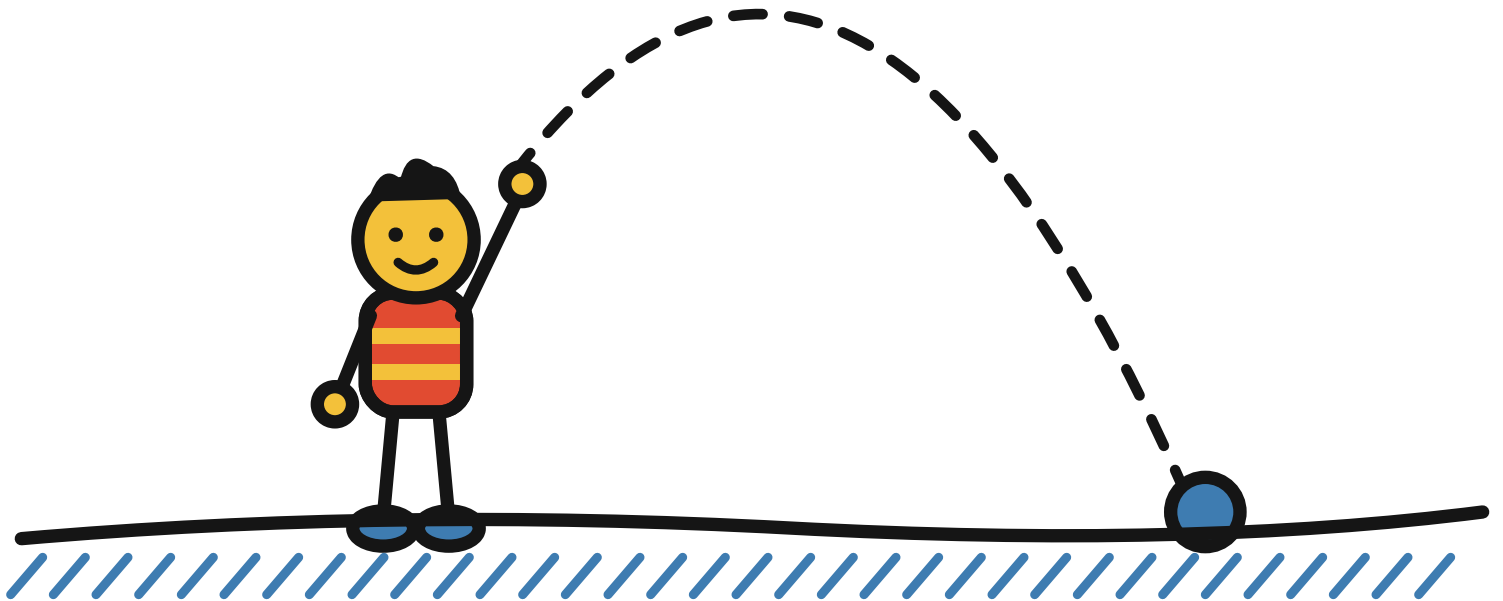
FALLING AND MISSING

Book Four · Why going up is only the easy half



Throw a ball.

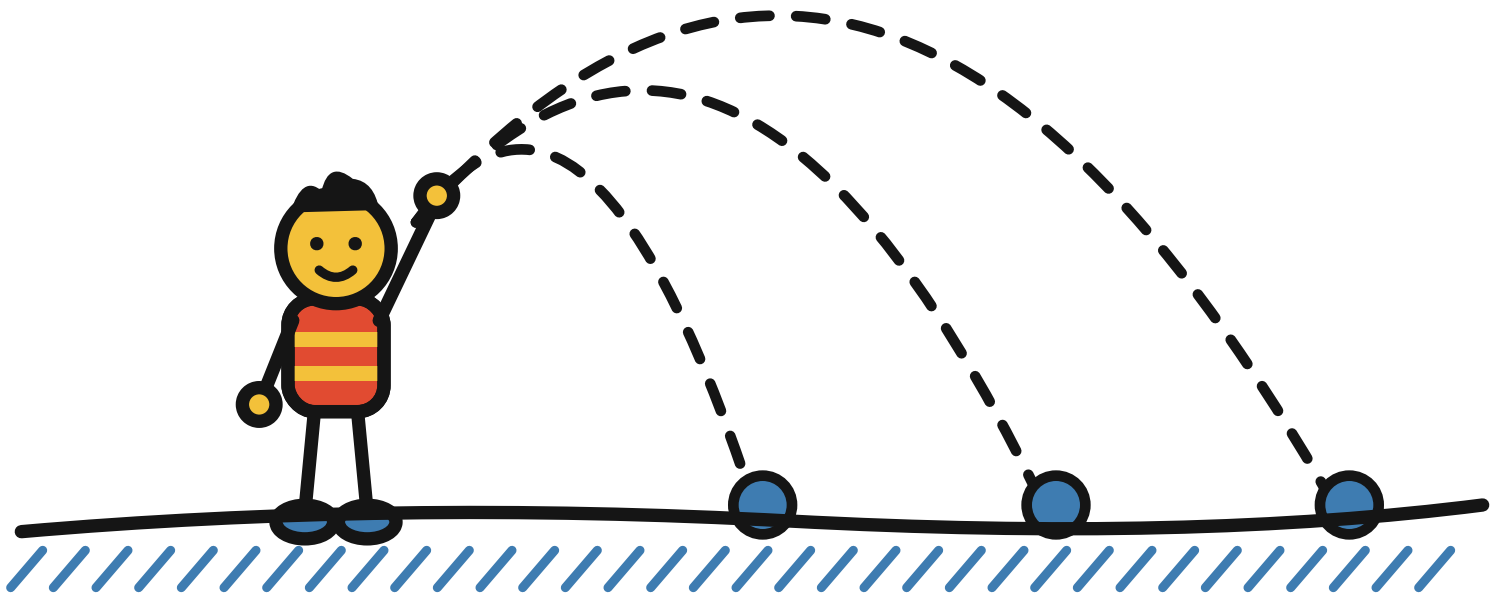
It goes up. Then it comes down.



It always comes down. Always.

Throw it harder.

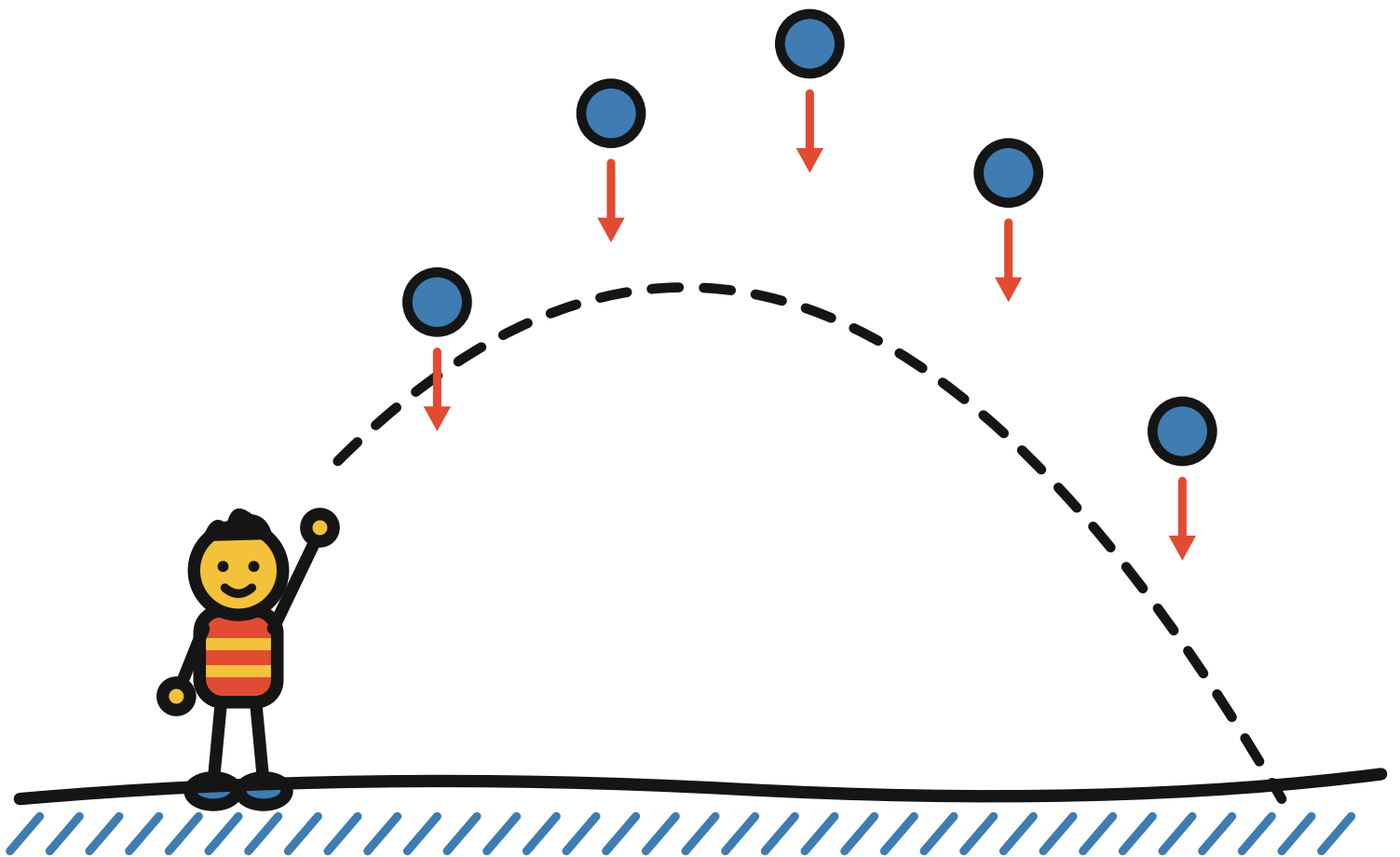
It lands further away.



Further. But it still lands.

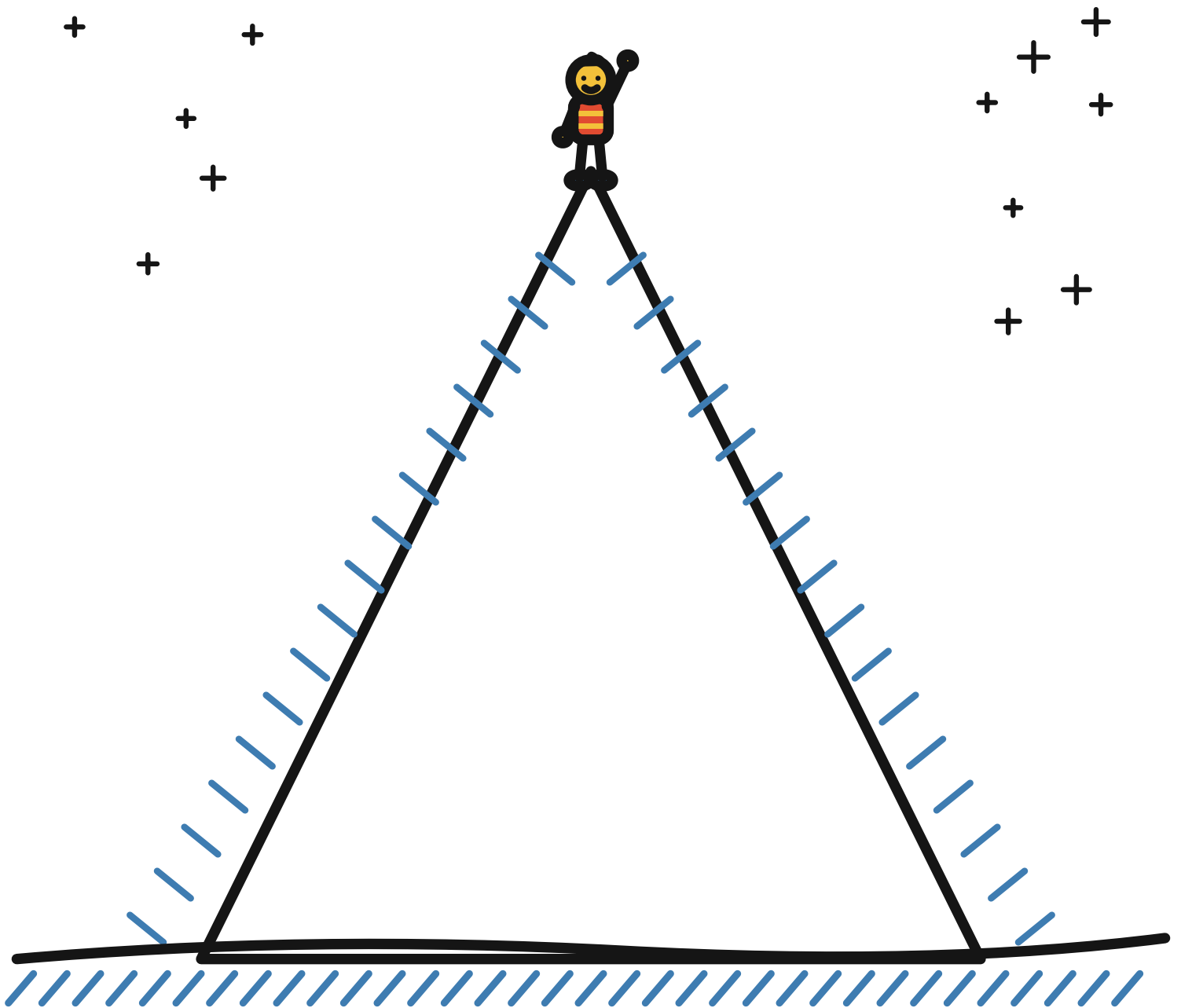
The ball is falling the whole time.

gravity is pulling it down at every moment



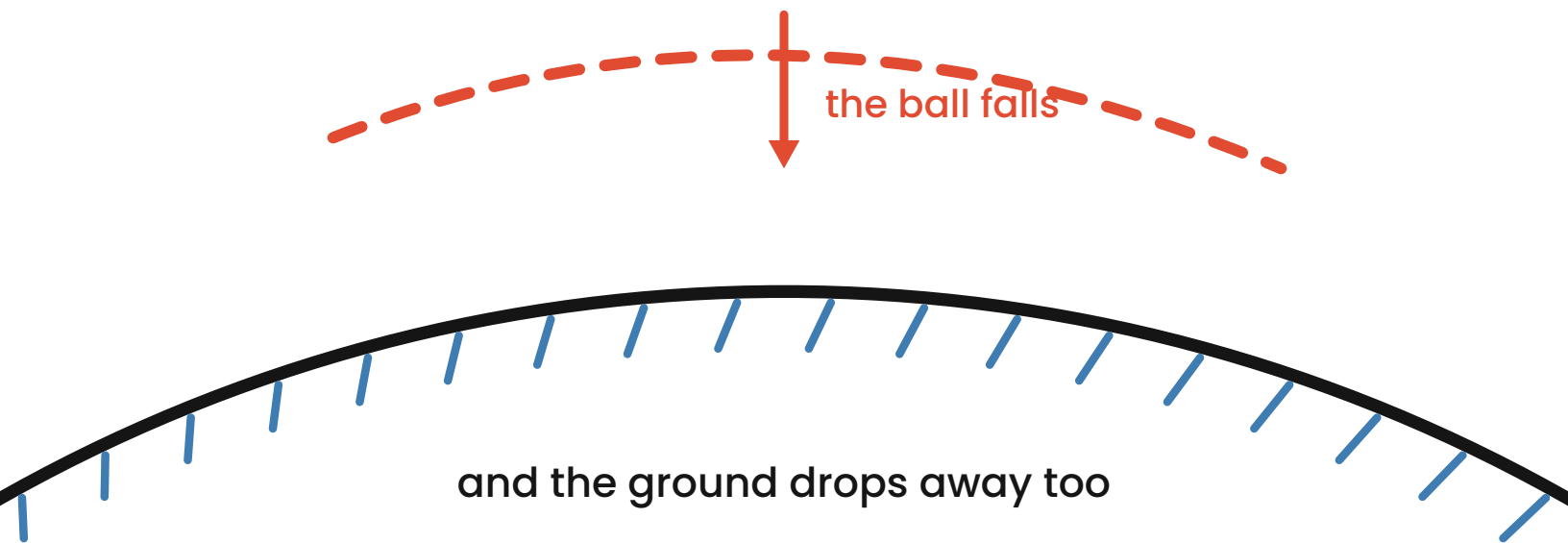
It is falling even while it is going up.

Now climb a very
tall mountain.



So tall that there is no air left to slow the ball.

**The ground curves
away underneath.**

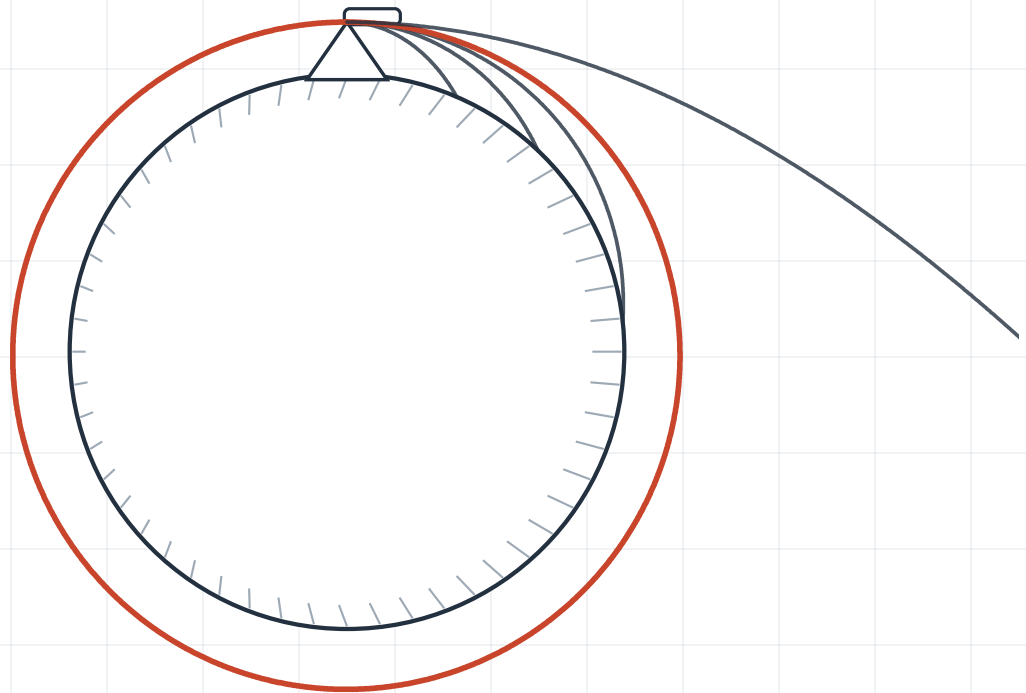


So the ball never quite gets there.

Throw it harder and harder.

Falling and Missing · figure 1 · after Newton, 1687

- 4.0 km/s falls back
- 5.6 km/s falls further
- 6.6 km/s further still
- 7.3 km/s never lands
- 10.5 km/s it leaves

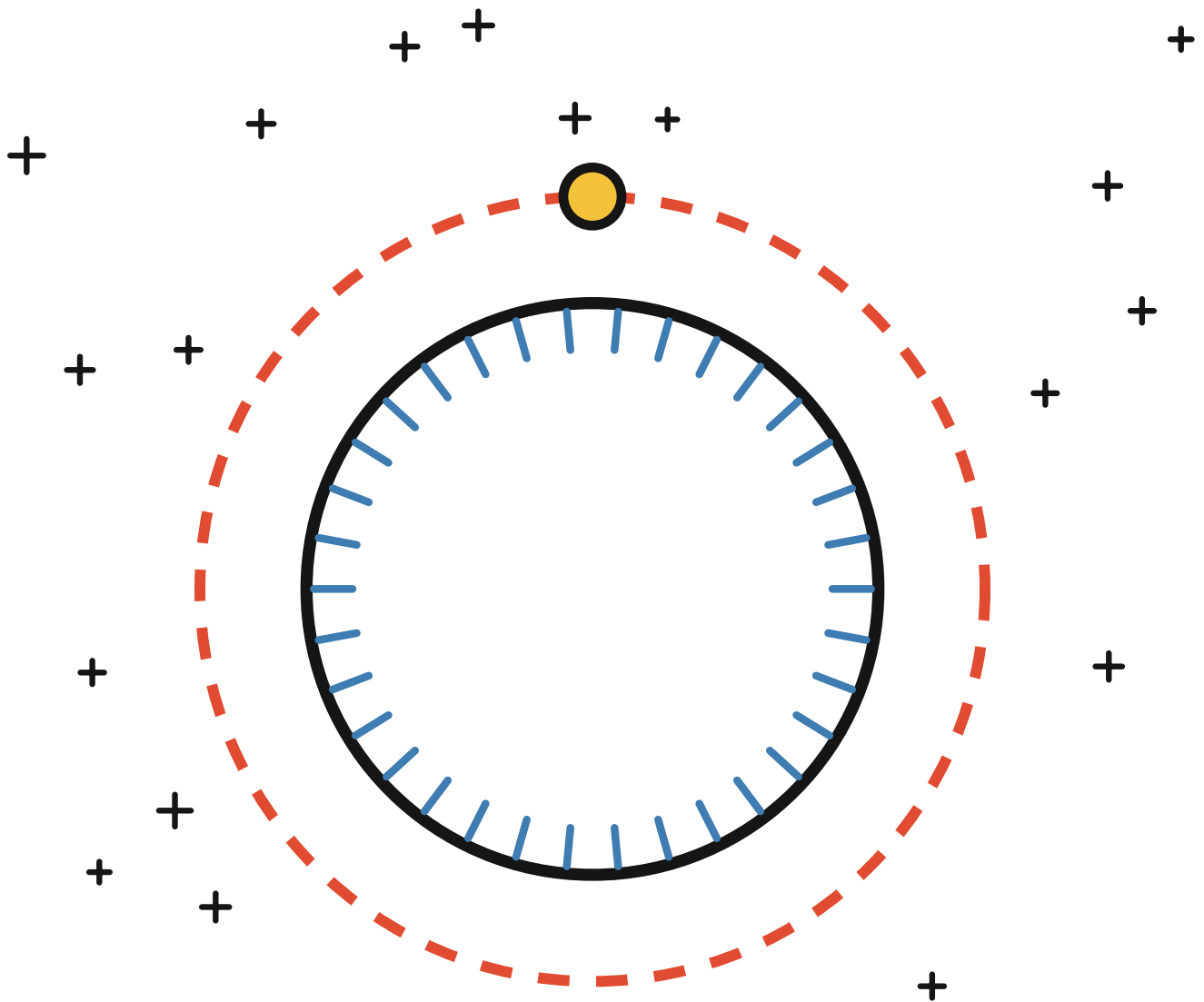


Every one of them is falling. Only one of them keeps missing.

The mountain is drawn far too tall on purpose, so the paths can be seen. Newton exaggerated his too.

Newton drew this in 1687. These curves are not sketched — they are computed from his own law of gravity.

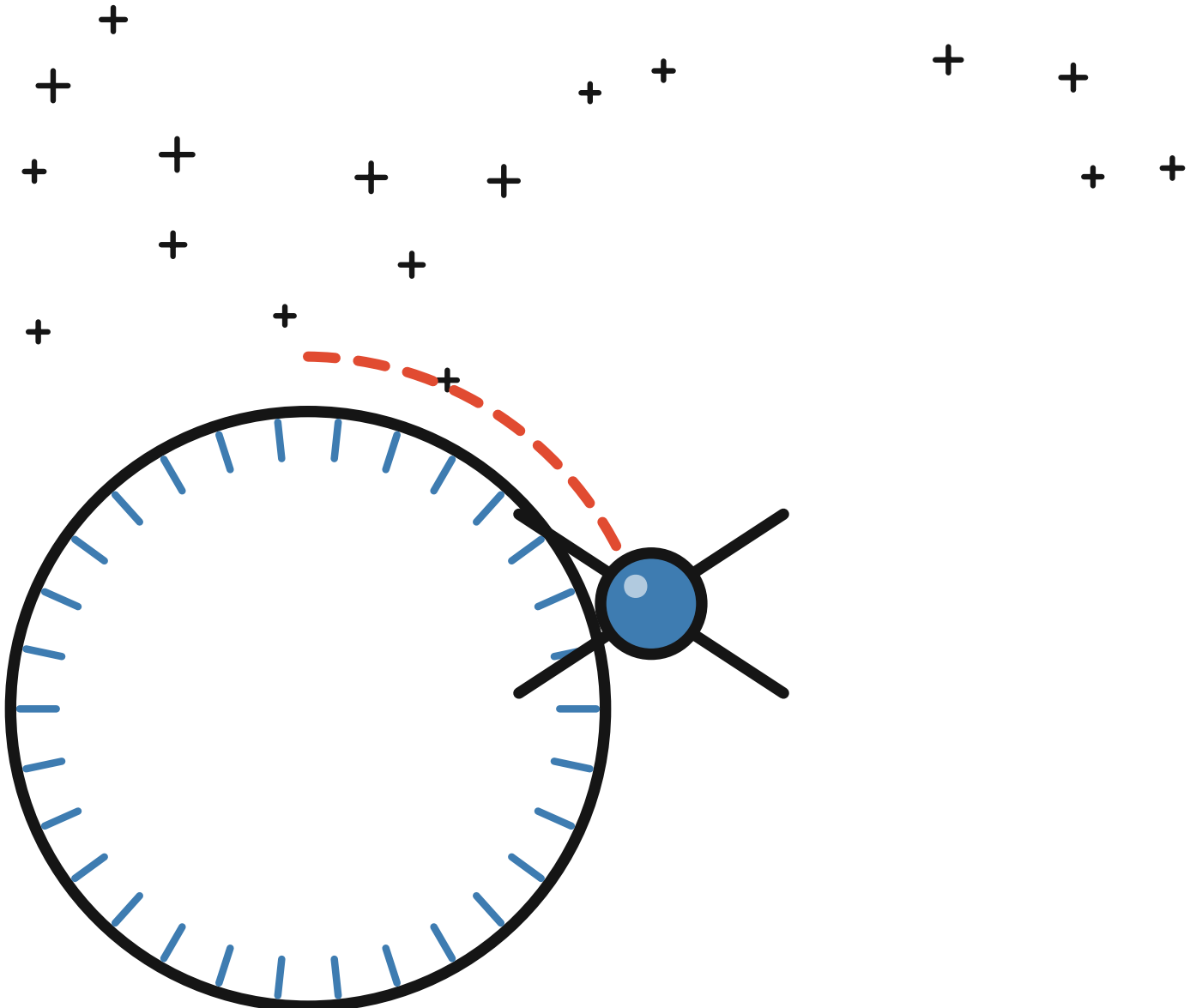
**Throw it hard enough
and you always miss.**



Round and round, falling the whole way.

That is all an orbit is.

Falling, and missing, for ever.



Going up is easy.
Staying up is not.



Almost all the work is in going sideways fast.

How fast is fast?

Eight kilometres. Every second.

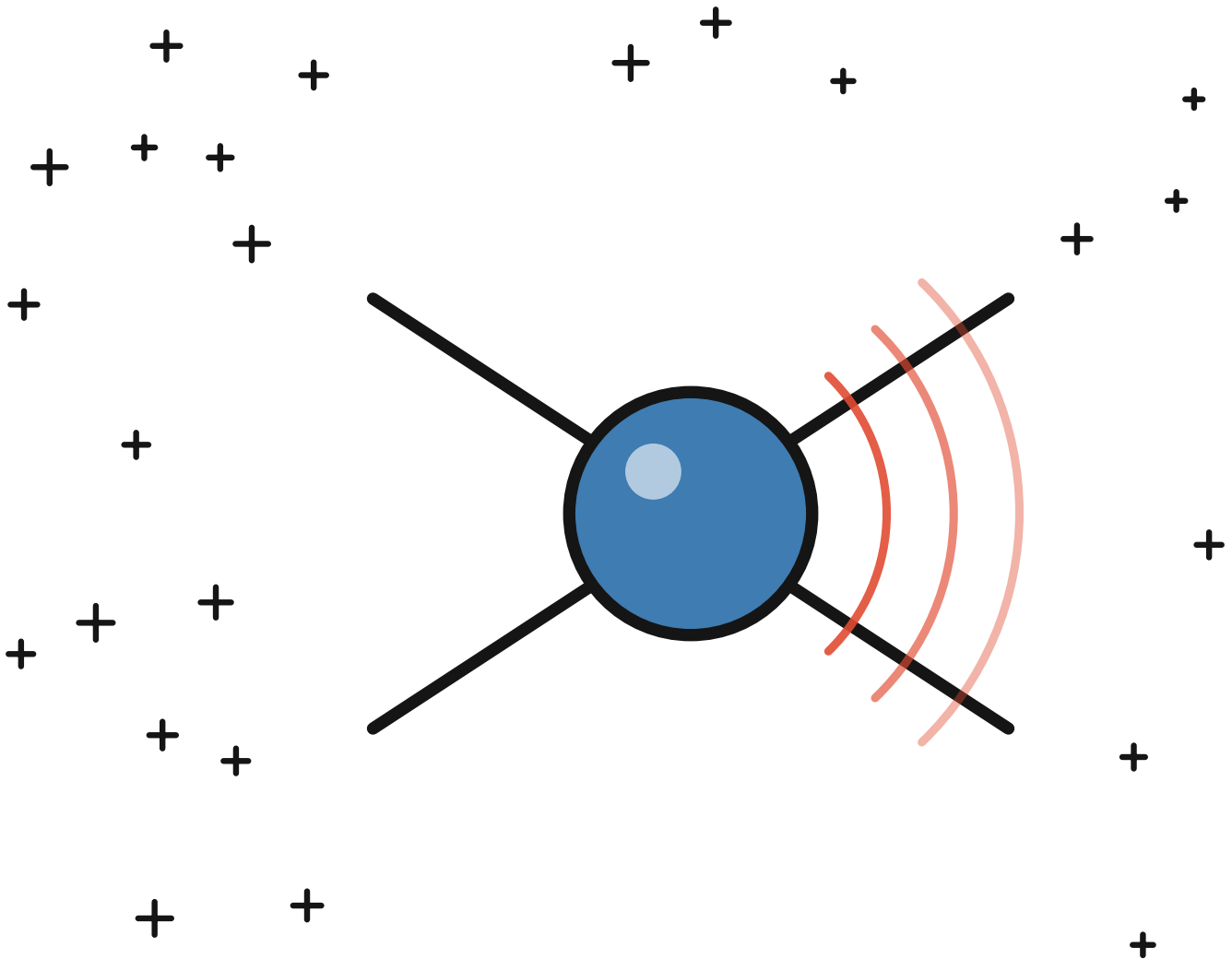


a fast car
would take four hours
to go that far



In one second, it crosses your whole town.

The first one to do it
was a beeping ball.

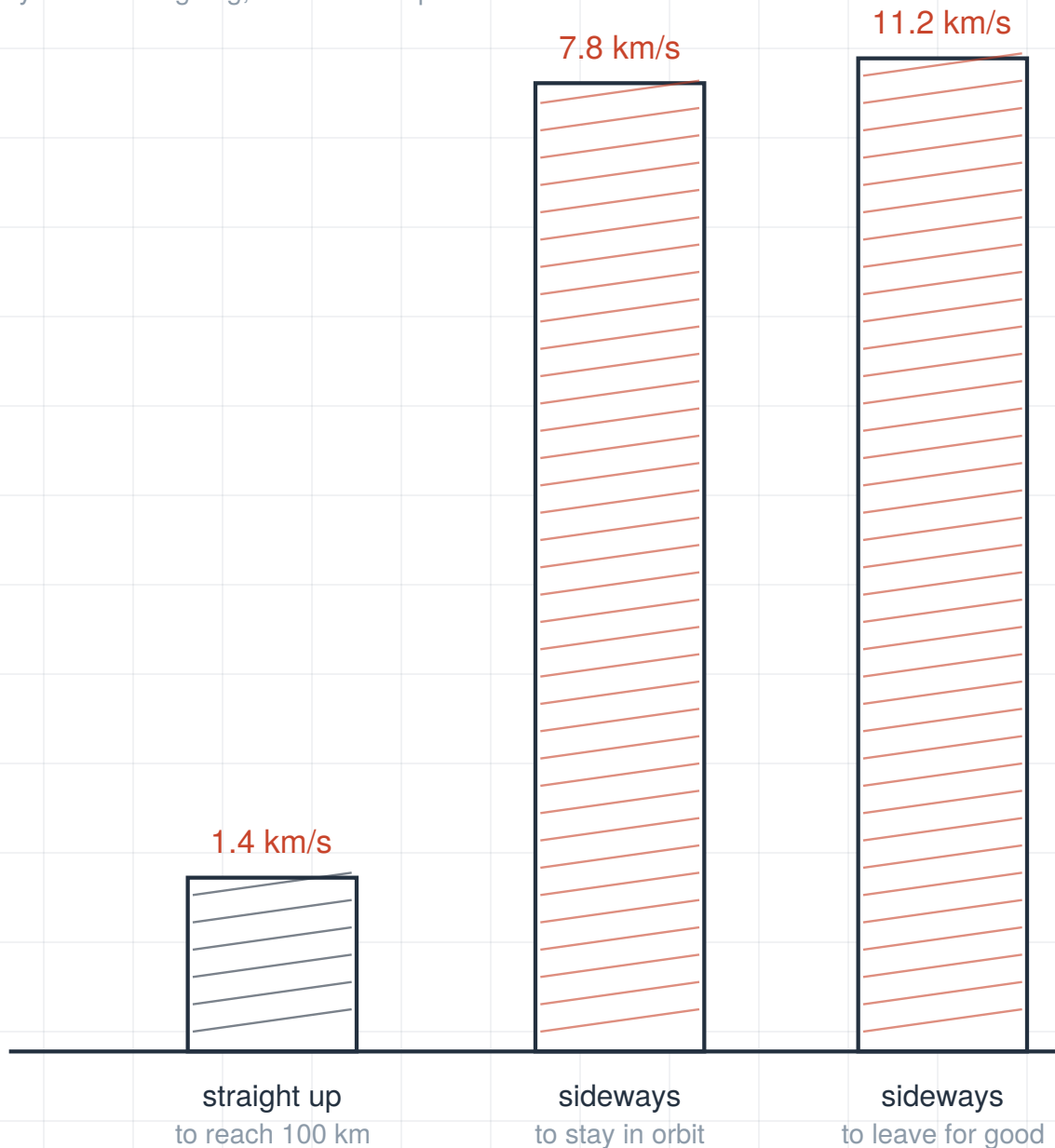


It went round the Earth once every hour and a half,
and it beeped the whole way.

The two speeds.

Falling and Missing · figure 2 · up versus sideways

how fast you must be going, in kilometres per second



Reaching space needs about 1.4 km/s. Staying there needs 7.8.
Energy goes as the square of speed, so orbit costs roughly
thirty times as much as simply getting up there.

Try this tonight.

1 Whirl something on a string.

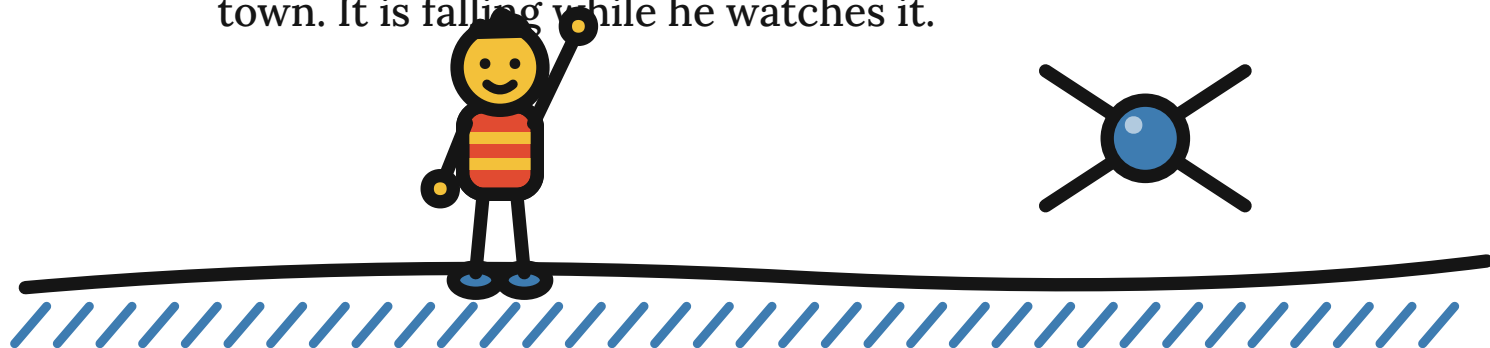
Tie a soft toy to a string and swing it round your head. Feel the pull in your hand — that is gravity's job. Let go and it flies off straight. Nothing is holding it in the circle except that pull.

2 Throw a ball off a step.

Drop one ball straight down and throw another sideways at the same moment. They hit the floor together. Going sideways does not slow the falling — that is the secret of the whole book.

3 Find the space station.

It is genuinely visible, bright and moving fast, a few evenings a month. Look up a pass time for your town. It is falling while he watches it.



For the grown-up.

The one idea

An orbit is not a place and it is not the absence of gravity. It is a trajectory. The object is in free fall the entire time, moving sideways fast enough that the curve of its fall matches the curve of the Earth. Astronauts float because they are falling, not because gravity has stopped.

Newton's cannonball

He drew it himself, in *A Treatise of the System of the World*, published after his death. A cannon on a mountain above the atmosphere, fired ever harder, until one trajectory closes into a circle. It remains the best explanation of orbit ever made, and the curves on page seven were computed from his own law rather than sketched.

The number that surprises people

Reaching 100 km straight up takes about 1.4 km/s. Staying there takes 7.8 km/s sideways, and escaping Earth takes 11.2. Since energy scales with the square of speed, orbit costs roughly thirty times what a hop to the same altitude does. This is why suborbital flights and orbital launches are entirely different problems.

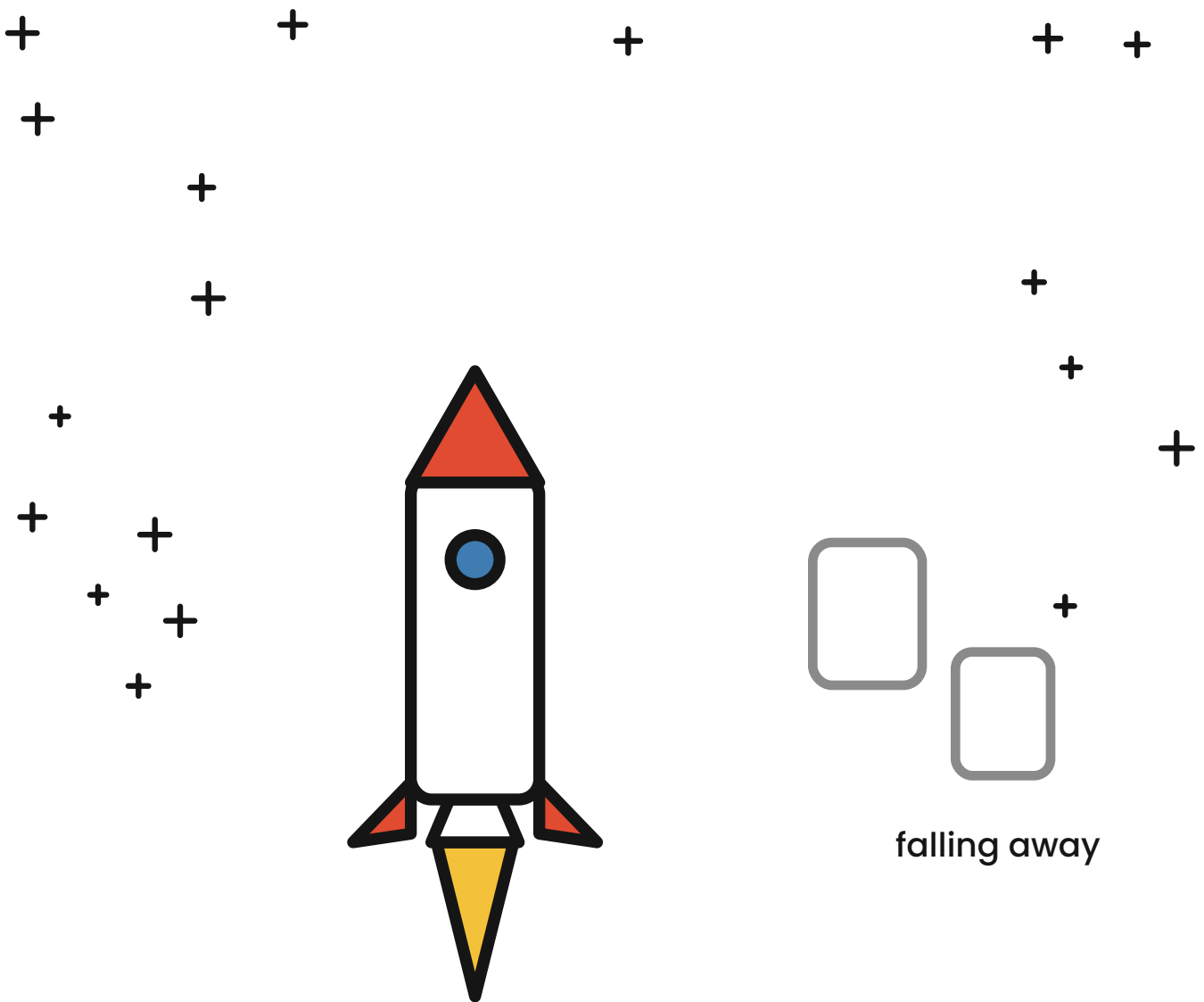
Sputnik

Launched 4 October 1957: a polished sphere 58 cm across, circling every 96 minutes at about 8 km/s. Its radio beep was the point — anyone with a shortwave set could confirm for themselves that something human was staying up.

Next:

Too Heavy to Lift

Book Five · why a rocket throws itself away



Fuel is heavy. So you need fuel
to lift the fuel.